AI-Powered Social Engineering Attack Predictor

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Problem Statement

Social engineering attacks are increasing because attackers use manipulative language, fake urgency, and emotionally persuasive messages to trick users into taking harmful actions. These threats are difficult to detect as they target human behavior rather than technical vulnerabilities. Most existing security tools only look for malware, suspicious links, or system-level anomalies and cannot analyze the linguistic or psychological manipulation present in text-based attacks.

As a result, users often fail to recognize deceptive messages and unknowingly fall for scams, phishing attempts, or identity theft. There is a critical need for an intelligent system that can process text, understand hidden manipulation cues, and accurately predict when a message is potentially dangerous. Such a system should guide users by providing early warnings, reducing the risk of human error, and strengthening overall cybersecurity awareness.

Abstract

Social engineering attacks use manipulative language and emotional tricks to deceive users into making harmful decisions. These attacks are difficult to detect because they target human behaviour rather than technical weaknesses. Most existing security systems are unable to understand the hidden intent behind such messages.

This project presents an AI-based system that analyses text messages to detect social engineering attacks. By using advanced models, especially a fine-tuned BERT model, the system understands message context and intent to provide accurate predictions. The proposed approach helps users identify risky messages early, reduce human error, and improve cybersecurity awareness.

Objectives

- Detect social engineering attacks using artificial intelligence
- Analyse message intent and manipulation cues
- Classify messages accurately as legitimate or malicious
- Select the best-performing detection model
- Reduce human error and enhance security awareness

Literature Survey (1/4)

S. No	Paper Title	Dataset	Techniques Used	Key Limitations
1	Social Engineering Attacks: A Survey — Albladi et al., 2019	Public research datasets, incident reports	Taxonomy-based classification, behavioral modelling	No AI-driven detection mechanism implemented
2	Fake Review Detection Using Deep Learning — Yang et al., 2024	Amazon and Yelp benchmark review datasets	LSTM + BERT hybrid deep learning model	Domain restricted to review fraud
3	Social Engineering Detection Techniques: Survey Study — Alghamdi et al., 2020	Public phishing and OSN datasets	Classical ML classifiers (SVM, NB, RF)	Limited dataset diversity and scalability
4	Detection of Fake Reviews Using Machine Learning Algorithm — Patel et al., 2020	Retail review dataset (limited size)	SVM, Random Forest, Logistic Regression	Dataset scale insufficient for generalization

Literature Survey (2/4)

S. No	Paper Title	Dataset	Techniques Used	Key Limitations
5	A Survey of Social Engineering Attacks — Alqarawi et al., 2021	Published cyber attack case studies	Attack vector categorization, tool mapping	Lacks defensive or preventive framework
6	Research on False Review Detection Methods — Lu et al., 2022	Review datasets from e-commerce platforms	Comparative analysis of supervised DL models	No real-time deployment or evaluation
7	The Shadow of Fraud: AI-powered Social Engineering — Sharma et al., 2024	AI-generated SE attack corpora	LLM-based prompt manipulation, adversarial modelling	Focused only on attack generation, not mitigation
8	Stay On-Topic: Generating Context-specific Fake Reviews — Yao et al., 2018	Large-scale hotel/restaurant review datasets	GAN-based text generation with topic control	Generates attacks instead of detecting them

Literature Survey (3/4)

S. No	Paper Title	Dataset	Techniques Used	Key Limitations
9	Fake Reviews Detection via Linguistic Features — Alam et al., 2020	Multi-domain opinion datasets	Stylometric linguistic feature extraction + ML	Language and writing-style dependent
10	Fact or Factitious? Opinion Spam Detection — Li et al., 2021	Review spam benchmark datasets	Contextual word embeddings (BERT-based)	High computational overhead
11	Password Strength Detection via ML — Liu et al., 2025	Leaked password datasets (e.g., RockYou)	ML-based password strength classification	Does not evaluate adversarial robustness
12	ML-Based Password Strength Estimation Approach — Zafar et al., 2023	Real-world leaked password sets	Feature-engineered ML pipeline	No protection against guessing attacks

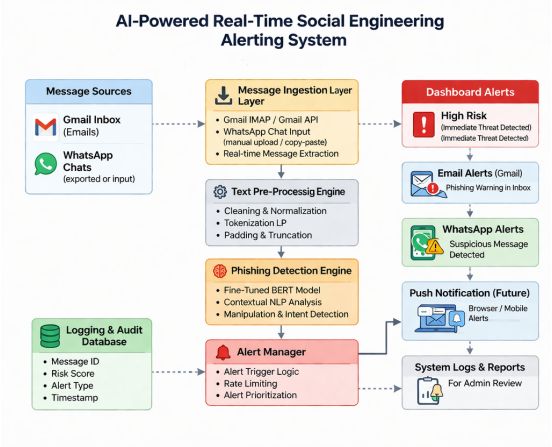
Literature Survey (4/4)

S. No	Paper Title	Dataset	Techniques Used	Key Limitations
13	AI-Driven Password Security Management System — Mehta et al., 2024	System design study (no dataset used)	Modular AI-driven security architecture	No empirical validation or deployment
14	AI in Password Security: Predicting Credential Attacks — Hossain et al., 2024	Credential breach datasets	AI-based attack probability prediction	No integrated security framework
15	Adversarial ML for Password Strength Estimation — Rao et al., 2025	Synthetic adversarial password datasets	Adversarial attack simulation	Identifies weaknesses but does not propose remediation

Data Overview

- Multiple text-based datasets were collected from different sources
- Datasets include Email and Chat messages
- Both legitimate and social engineering attack messages are present
- Only relevant columns such as message text and labels were selected
- Different datasets were merged into a single unified dataset
- Labels were standardized into two classes: Legitimate and Attack
- The final cleaned dataset was used for training and evaluation

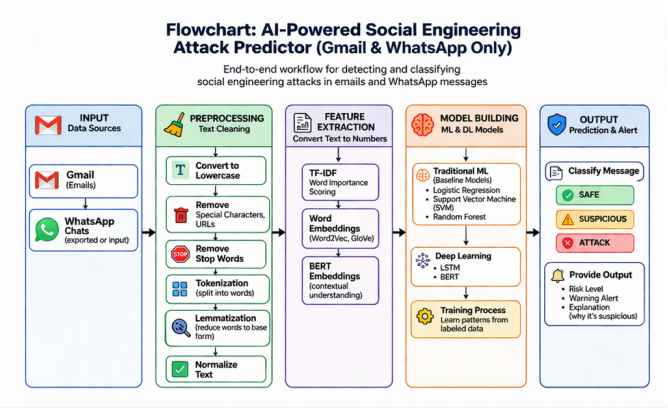
System Architecture



System Architecture

- The system collects incoming emails and whatsapp messages from user accounts.
- Messages are securely fetched using Gmail API and messaging service APIs.
- The message content is cleaned, tokenized, and prepared for analysis.
- A BERT-based NLP model analyzes the text to identify phishing and social engineering patterns.
- Each message is assigned a risk score based on its content.
- When the risk exceeds a predefined threshold, alerts are generated via dashboard notifications, email, or whatsapp messages.
- All message details, predictions, and alerts are stored in a database for monitoring and further analysis.

Flowchart


Flowchart: AI-Powered Social Engineering Attack Predictor (Gmail & WhatsApp Only)
End-to-end workflow for detecting and classifying social engineering attacks in emails and WhatsApp messages
INPUT
Data Sources
Gmail
(Emails)
WhatsApp Chats
(exported or input)
PREPROCESSING
Text Cleaning
Convert to Lowercase
Remove Special Characters, URLs
Remove Stop Words
Tokenization
(split into words)
Lemmatization
(reduce words to base form)
Normalize Text
FEATURE EXTRACTION
Convert Text to Numbers
TF-IDF
Word Importance Scoring
Word Embeddings
(Word2Vec, GloVe)
BERT Embeddings
(contextual understanding)
MODEL BUILDING
ML & DL Models
Traditional ML
(Baseline Models)
• Logistic Regression
• Support Vector Machine (SVM)
• Random Forest
Deep Learning
• LSTM
• BERT
Training Process
Learn patterns from labeled data
OUTPUT
Prediction & Alert
Classify Message
SAFE
SUSPICIOUS
ATTACK
Provide Output
• Risk Level
• Warning Alert
• Explanation
(why it's suspicious)

Methodology

- Data Collection
- Data Integration
- Text Preprocessing
- Feature Extraction
- Model Training
- Model Evaluation
- Model Selection
- Deployment

Execution

- **Data Collection**
  - Email and Chat message datasets
  - Includes both legitimate and attack messages
- **Data Preparation**
  - Combined all datasets into a unified dataset
  - Text cleaning and label normalization
- **Feature Extraction**
  - TF-IDF for machine learning models
  - Tokenization for deep learning and BERT
- **Model Training**
  - ML Models: Logistic Regression, SVM, Random Forest
  - DL Models: LSTM and GRU
  - Transformer Model: Fine-tuned BERT

Execution (contd..)

- **Model Evaluation**
  - Accuracy, Precision, Recall, and F1-Score
- **Model Deployment**
  - Best-performing model (BERT) selected
  - Real-time detection using Streamlit

Execution (contd..)

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	98.03%	97.87%	96.57%	97.21%
SVM	98.82%	98.68%	97.99%	98.34%
Random Forest	99.72%	99.71%	99.49%	99.61%
LSTM	98.53%	98.10%	97.77%	97.94%
GRU	98.83%	98.46%	98.26%	98.36%
BERT	99.27%	99.19%	98.76%	**98.98%**
BERT	**99.27%**	**99.19%**	98.76%	**98.98%**

Figure 1: Evaluation Metrics

Execution (contd..)

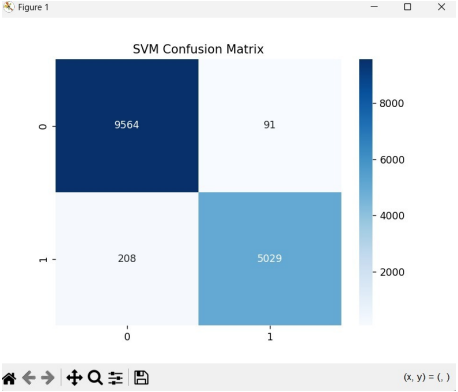


Figure 2: Confusion Matrix

Conclusion

Developed an AI-based system to detect social engineering attacks using Natural Language Processing (NLP) and Machine Learning (ML), enabling the identification of manipulative and deceptive text that often bypasses traditional security methods. The system analyzes communication patterns and linguistic cues to accurately recognize potential threats, providing real-time predictions and user alerts to prevent cyber attacks. By proactively detecting such vulnerabilities, it not only strengthens security measures but also enhances user awareness, contributing to improved overall cybersecurity protection.

Results

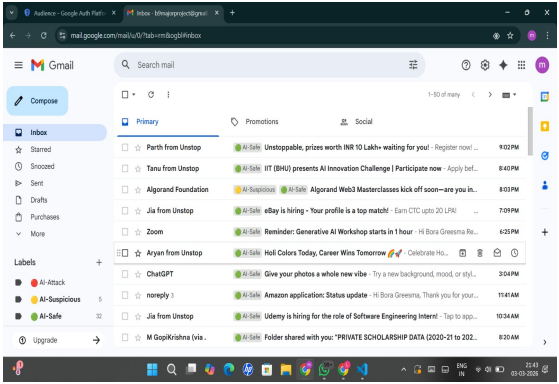


Figure 3: Gmail-Based Phishing Scam

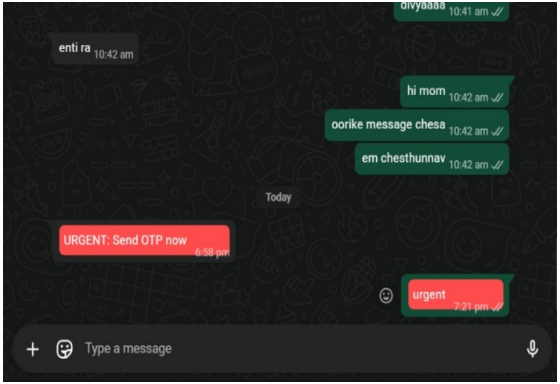


Figure 4: WhatsApp OTP Phishing Message

Results (contd..)

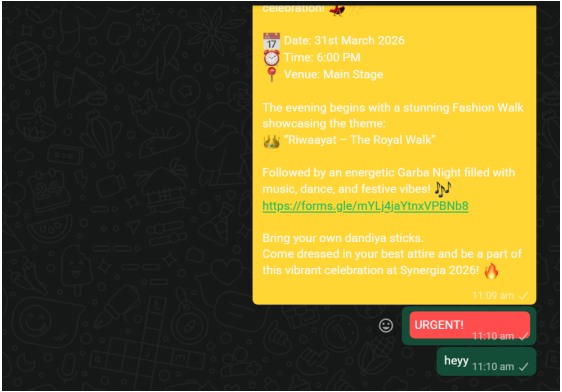


Figure 5: WhatsApp OTP Phishing Message

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Thank You!